

Entangled or not Entangled Q-Bits - Cheat Sheet

To determine whether the quantum state of the example $|b\rangle = \frac{1}{\sqrt{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle + \frac{1}{\sqrt{3}}|10\rangle$ is **entangled** or **separable**, we check if it can be written as a **tensor product** of two single-qubit states:

$$|b\rangle = |\psi\rangle \otimes |\phi\rangle$$

Step 1

it is a two-qubit system, with basis:

$|00\rangle, |01\rangle, |10\rangle, |11\rangle$ (this is always the same)

example: $|b\rangle = \frac{1}{\sqrt{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle + \frac{1}{\sqrt{3}}|10\rangle$

Note: There is **no component of** $|11\rangle$, i.e., its amplitude is 0.

Step 2

Assume:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad |\phi\rangle = c|0\rangle + d|1\rangle$$

Then the product is:

$$|\psi\rangle \otimes |\phi\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

Compare to:

$$|b\rangle = \frac{1}{\sqrt{3}}|00\rangle + \frac{1}{\sqrt{3}}|01\rangle + \frac{1}{\sqrt{3}}|10\rangle + 0 \cdot |11\rangle$$

This matches to:

- $ac = \frac{1}{\sqrt{3}}$
- $ad = \frac{1}{\sqrt{3}}$
- $bc = \frac{1}{\sqrt{3}}$
- $bd = 0$

Since $b \times d = 0$ it means, that,

either $b = 0$

or $d = 0$

if $b = 0$:

$b \times c = 0$, but $bc = \frac{1}{\sqrt{3}} \neq 0 \rightarrow$ contradiction.

If $d = 0$:

$a \times d = 0$, but $ad = \frac{1}{\sqrt{3}} \rightarrow$ contradiction

No solution exists:

$\Rightarrow$ cannot be written as a product state.

⇒ The state **cannot** be separated into two independent qubits
⇒ the state is **entangled**

Schmidt Decomposition

Schmidt decomposition:

$$|\psi\rangle = \alpha|00\rangle + \beta|01\rangle + \gamma|10\rangle + \delta|11\rangle$$

Form a 2×2 matrix:

$$M = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$$

Compute the **determinant**:

$$\det(M) = \alpha\delta - \beta\gamma$$

- If $\det(M) = 0$, the state is **not entangled** (separable).
- If $\det(M) \neq 0$, the state **is entangled**.